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|  | Document Type:  | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Document Code  | <b>RES- RP-01</b>        |
|                                                                                   | Document Title: | <b>RESEARCH PROPOSAL</b>                       | Revision No.   | <b>00</b>                |
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|                                                                                                                                        |                                   |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>(1) Research Program/Project/Study Title:</b> HANDMADE PAPER MAKING FROM FALLEN LEAVES AND STEM OF SELECTED LOCALLY AVAILABLE TREES |                                   |
| <b>Proponent Name:</b>                                                                                                                 | <b>Rhodora Ticar</b>              |
| <b>Academic Rank:</b>                                                                                                                  | <b>Asso. Professor V</b>          |
| <b>Co-Authors (with academic rank):</b>                                                                                                |                                   |
| <b>Campus/Satellite College/Address:</b>                                                                                               | <b>Mambusao Satellite College</b> |
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

|                          |  |
|--------------------------|--|
| <b>Component Title:</b>  |  |
| <b>Component Leader:</b> |  |
| <b>Academic Rank:</b>    |  |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(2) Category (please check)</b><br><input type="checkbox"/> Social Science/Education /Humanities & other related fields<br><input checked="" type="checkbox"/> Natural/Biological Sciences<br><input type="checkbox"/> Food Research/<br><input type="checkbox"/> Developmental/ Computer/Engineering/ Architecture<br><input type="checkbox"/> Extension<br><input type="checkbox"/> Poster | <b>(3) Research Center</b><br>[Check the concerned center]<br><br><input type="checkbox"/> SSITRDC<br>Social Science and Industrial Technology Research Development Center<br><br><input type="checkbox"/> LRDC<br>Livestock Research Development Center<br><br><input type="checkbox"/> FFRDC<br>Fisheries and Food Research Development Center<br><br><input type="checkbox"/> CSRDC<br>Crop Science Research Development Center |
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|                                   |                 |
|-----------------------------------|-----------------|
| <b>(4) Site of implementation</b> |                 |
| Municipality/City:                | <b>Mambusao</b> |
| Province:                         | <b>Capiz</b>    |
| Region:                           | <b>VI</b>       |

|                                                                                                                                                                                                                                                                                                                                                            |                                                            |                     |                                |                                        |                                             |                                                            |                                                                                                                                                                                                                                                                                                                   |
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| <b>(5) Classification</b><br><br><table> <tr> <td><b>Research:</b></td> <td><b>Development:</b></td> </tr> <tr> <td><input type="checkbox"/> Basic</td> <td><input type="checkbox"/> Pilot Testing</td> </tr> <tr> <td><input checked="" type="checkbox"/> Applied</td> <td><input type="checkbox"/> Tech. Promotion/Commercialization</td> </tr> </table> | <b>Research:</b>                                           | <b>Development:</b> | <input type="checkbox"/> Basic | <input type="checkbox"/> Pilot Testing | <input checked="" type="checkbox"/> Applied | <input type="checkbox"/> Tech. Promotion/Commercialization | <b>(6) Mode of Implementation</b><br><br><input checked="" type="checkbox"/> Single Authorship/Proponent<br><input type="checkbox"/> Collaborative Research (within campus)<br><input type="checkbox"/> Collaborative Research (intercampus)<br><input type="checkbox"/> Collaborative Research (external agency) |
| <b>Research:</b>                                                                                                                                                                                                                                                                                                                                           | <b>Development:</b>                                        |                     |                                |                                        |                                             |                                                            |                                                                                                                                                                                                                                                                                                                   |
| <input type="checkbox"/> Basic                                                                                                                                                                                                                                                                                                                             | <input type="checkbox"/> Pilot Testing                     |                     |                                |                                        |                                             |                                                            |                                                                                                                                                                                                                                                                                                                   |
| <input checked="" type="checkbox"/> Applied                                                                                                                                                                                                                                                                                                                | <input type="checkbox"/> Tech. Promotion/Commercialization |                     |                                |                                        |                                             |                                                            |                                                                                                                                                                                                                                                                                                                   |

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| <b>(7) Sector/Commodity</b><br><b>Natural Science</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>(8) Discipline/Field of Specialization</b><br>Gen. Science |
| <p><b>(9) Rationale</b> (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R &amp; D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)</p> <p>Proper utilization of these resources (available local resources) may generate sufficient employment opportunities there leading to economic betterment of rural people. Agricultural resource, horticultural resource, forest resource, livestock resource etc. could be utilized profitably in rural periphery. During harvest time large amount of vegetables and fruits go into wastage or sold in a throw away price. If these are processed in rural level, along with generation of employment opportunities, wastage of different products will be minimize and farmers will also get the fair price for their production (Sukul, 2017).</p> <p>The growing environmental problems, the problem of waste disposal and the depletion of non-renewable resources have stimulated the use of green materials compatible with the environment to reduce environmental impacts. Therefore, there is a need to design products by using natural resources. Natural fibers seem to be a good alternative since they are abundantly available and there are a number of possibilities to use all the components of a fiber-yielding crop; one such fiber-yielding plant is Agave Americana. The leaves of this plant yield fibers and all the parts of this plant can be utilized in many applications. The “zero-waste” utilization of the plant would enable its production and processing to be translated into a viable and sustainable industry (Hulle et al 2015).</p> <p>Nowadays, the development of environmentally friendly products is at its core point, in order to the constant study for practical solutions to environmental problems and the urgent need to change from non-renewable products. Multiple investigations have been developed in this regard trying to find new materials in order to replace the conventional ones such as plant fibres which have so many advantages like renewable natural resource, biodegradable, cheap skate in cost, good specific mechanical properties and low density (Wolela, 2019).</p> <p>Paper making from leaves and stem of plants,as one of the ways to utilize locally available resources and probably provide employment opportunitites, were known in the scientific world for many decades. Aside from the plant mentioned above ( Agave Americana), suitability of aquatic plant fibers for handmade papermaking was studied by Bidin et al (2015). Extraction and characterization of natural cellulose fibers from Snake plant, <i>Sanseveria trifasciata</i> was also studied through its leaves (Wolela, 2019).</p> <p>Handmade paper industry is an environmental friendly and very promising industry for local entrepreneurship.Papers made from aquatic plants have multiple uses.They can be used for writing, food wrapping, tissue paper, and book mark and can be commercialized as value added materials for handmade crafts (Bidin et al, 2015).</p> <p>Borrowing the statement of of Obluska (2019) in her article entitled” Paper from fallen leaves – old idea, new approach”, she stated “<i>Contrary to what is going through the media, paper from fallen leaves doesn’t look like a white sheet we usually put into the printer. It will not be used in such purpose either. Cellulose obtained from the leaves will be used for cardboard, wrapping paper and paper pulp packaging production. What’s more – it was decided that the leaves will constitute not 100, but 75% of the final product. The rest will be traditional cardboard. Nevertheless, paper from fallen leaves, even if not entirely from lawn’s litter, is a step forward. It will definitely improve air quality in Ukrainian cities that would otherwise be polluted by smoke from burning leaves</i>”.</p> <p>With all of these, this study is hereby conceptualized.</p> |                                                               |
| <p><b>(9-A) Project Description</b> (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                               |

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relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

1. Identification and Selection of Locally Available trees
2. Collection of Materials (stems and leaves) from the Identified Trees
3. Conduct of steps for paper making
4. Evaluation of the product in terms of texture and acceptability
5. Measuring the density, thickness and puncture resistance of the product

**(10) Objectives/Statement of the Problem** (for Social Sciences ) (Statement of the specific purpose to address the problem areas of the project.)

#### Objectives of the Study

1. To be able to produce and evaluate the quality of paper made from *varying levels* of fallen leaves of selected locally available trees in terms of texture and general acceptability.
2. To be able to produce and evaluate the quality of paper made from fallen small stems of selected locally available trees in terms of texture and general acceptability.
3. To produce and evaluate paper from the combined mixture of leaves of two locally available trees.
4. To determine the density, thickness and puncture resistance of the paper made from fallen leaves and stems of selected locally available trees.

**(11) Review of Literature/Prior Art Search** (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Increasing concerns for future fiber supplies in pulp and paper industries has shifted interest in nonwood sources from agriculture residues and aquatic plants (Bidin et al, 2015).

Fiber dimensions and lignin and cellulose content of various highly productive, non-wood plants and agricultural residues were examined to assess their suitability for paper production. Plants like kenaf (*Hibiscus cannabinus* L.) and giant reed (*Arundo donax* L.) internodes gave very good derived values, especially slenderness ratio, which is directly comparable to some softwood and most hardwood species. Cotton (*Gossypium hirsutum* L.) stalks, miscanthus (*Miscanthus × giganteus*) and switchgrass (*Panicum virgatum* L.) have shorter fibers resulting to poorer flexibility and Runkel ratios, but still satisfactory slenderness ratios. Finally, fibers from olive tree (*Olea europea* L.) and almond tree (*Prunus dulcis* L.) prunings presented relatively short and thick fibers producing the poorest derived values among all the species examined. Fiber dimensions did not differ significantly within each species, when samples from different stalks/branches or different positions (base, middle top) were examined. The only exception were cotton stalks, where those differences did not have any significant effect on fiber derived values. Chemical analysis of the raw plant materials revealed satisfactory levels of -cellulose content (close to 40%) and Klason lignin content (<30%) compared to those of hardwoods and softwoods. Relatively increased (>25%) lignin content in miscanthus, switchgrass and almond prunings may require additional pulping time and chemical charge compared to those of other non-wood raw materials. Analysis of samples at various heights/lengths of the plant materials showed that lignin and cellulose content depends on tissue maturity, but does not change significantly within each species (Ververis, et al 2004).

The project obtained a patent for “the technology of recycling leaves with the subsequent production of paper” in 2019, and Research and Production Enterprise Re-Leaf Technology LLC was founded in 2021. Currently, the company has a production capability of up to 5,000 tonnes annually (Cole 2022).

“In 2020 he founded the RE-Leaf PAPER company in partnership with Andriy Vartsaba. They reached an agreement with Kyivzelenbud for the supply of fallen leaves from Kyiv and in October produced their first 1.5 tons of recycled industrial paper. It has a density of 90 and 100 g/m<sup>2</sup>, a thickness of 0.2 mm, and a puncture resistance of 285 kPa. The paper will be used to make samples of paper bags, utensils and cardboard “(bioukraine.org, 2020).

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Valentyn Frechka, a Ukrainian student transformed fallen leaves into paper (bioukraine.org/news, 2020). A patent application from South Korea was published online stating the manufacturing method for raw material of paper box using of fallen leaves and tobacco composite. She/he pulverize a mixture of multiple type of leaves, deinking a mixture of pulverize, punching the deinked leaves so as to form a composition of fallen leaves and drying of the waste of tobacco stem and pulverizing, etc.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

Cellulose, a complex carbohydrate, or polysaccharide, consisting of 3,000 or more glucose units. The basic structural component of plant cell walls, cellulose comprises about 33 percent of all vegetable matter (90 percent of cotton and 50 percent of wood are cellulose) and is the most abundant of all naturally occurring organic compounds. Nondigestible by humans, cellulose is a food for herbivorous animals (e.g., cows, horses) because they retain it long enough for digestion by microorganisms present in the alimentary tract; protozoans in the gut of insects such as termites also digest cellulose. Of great economic importance, cellulose is processed to produce papers and fibres and is chemically modified to yield substances used in the manufacture of such items as plastics, photographic films, and rayon. Other cellulose derivatives are used as adhesives, explosives, thickening agents for foods, and in moisture-proof coatings (<https://www.britannica.com/science/cellulose>).

Cellulose (C<sub>6</sub>H<sub>10</sub>O<sub>5</sub>)<sub>n</sub> is one of the most ubiquitous organic polymers on the planet. It is a significant structural component of the primary cell wall of green plants, various forms of algae and oomycetes. It is a polysaccharide consisting of a linear chain of several hundred to many thousands of β(1 → 4) linked d-glucose units. There are various extraction procedures for cellulose developed by using different processes like oxidation, etherification and esterification which convert the prepared celluloses in to cellulose derivatives. Since it is a non-toxic, bio-degradable polymer with high tensile and compressive strength, it has widespread use in various fields such as nanotechnology, pharmaceutical industry, food industry, cosmetics, textile and paper industry, drug-delivery systems in treating cancer and other diseases. Micro-crystalline cellulose in particular is among the most frequently used cellulose derivatives in the food, cosmetics, pharma industry, etc. and is an important excipient due to its binding and tableting properties, characterized by its plasticity and cohesiveness when wet. Bacterial cellulose's high dispensability, tasteless and odourless nature provides it with lot of industrial applications. Currently, about half of the waste produced in India contains about 50% cellulose which can be used productively (<https://www.intechopen.com/chapters/67083>).

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

**A) VARIABLES TO BE MEASURED AND EVALUATED**

1. texture
2. general acceptability
3. density
4. thickness

**B) TREATMENTS TO BE USED AND THEIR LAYOUT**

**Fallen Leaves Composition**

- 1- 100% Fallen Leaves of Mango Tree
- 2- 100% Fallen Leaves of Mahogany Tree
- 3- 100% Fallen Leaves of Lansones Tree
- 4- 50% Fallen Leaves of Mango Tree + 50% Fallen Leaves of Mahogany Tree
- 5- 50% Fallen Leaves of Mango Tree + 50% Fallen Leaves of Lansones Tree
- 6- 50% Fallen Leaves of Mahogany Tree+ 50% Fallen Leaves of Lansones Tree

**Fallen Stem/Twigs Composition**

- 1- 100% Fallen Stem/Twigs of Mango Tree
- 2- 100% Fallen Stem/Twigs of Mahogany Tree
- 3- 100% Fallen Stem/Twigs of Lansones Tree

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- 4- 50% Fallen Stem/Twigs of Mango Tree + 50%Fallen Stem/Twigs of Mahogany Tree
- 5- 50% Fallen Stem/Twigs of Mango Tree + 50%Fallen Stem/Twigs of Lansones Tree
- 6- 50% Fallen Stem/Twigs of Mahogany Tree+ 50%Fallen Stem/Twigs of Lansones Tree

#### C) EXPERIMENTAL PROCEDURES AND DESIGN

Experimental design will be used. Since the study will be conducted in the lab set-up, complete randomized design will be used with six treatments replicated three times

### PROCEDURE OF MAKING PAPER OUT OF FALLEN LEAVES

#### MATERIALS/FACILITIES

- a. 4 pair of scissors
- b. 1 kg Caustic soda (20 percent of the dry fiber weight; 3-1/2 ounces of washing soda and 8 quarts of water per pound of dry fiber)
- c. Water
- d. a large, nonfood, nonreactive pot (stainless steel, glass, or enamel-coated)
- e. weighing scale
- f. 4 pot holders
- g. 4 nonfood, nonreactive stirring utensils
- h. a mesh strainer
- i. 4 bucket
- j. 1 pack rubber gloves
- k. 2 nonfood blender
- l. 2 googles
- m. 20 pcs N95 face mask
- n. Mould and deckle.
- o. Vat. This is a tub larger than mould and deckle.
- p. Hard felt cloth (1mm x 9 inches x 12 inches)
- q. Plastic buckets with handles. These will hold pulp and help in draining vats. Papermaking uses a lot of water.
- r. A press or sponges and brayer.
- s. A flat work surface

#### PROCEDURE

1. Collect the fallen leaves and twigs from each tree
2. Dry using dehydration or through sundrying
3. weigh your dry fiber before wetting it. Weigh it according to the treatments given
4. With scissors, cut the fiber into 1/2-inch to 1-inch strips (if you're going to hand-beat fibers, cut them into 2-inch pieces) to reduce cooking time and prevent tangling in the blender. Sort out twigs or other foreign material.
5. Soak fiber in plain water overnight to fully hydrate before cooking and processing.
6. Fill a pot with water to cover the fiber, about 2 gallons per pound. Wear gloves, measure the washing soda or other alkali, 20 percent of your dry weight, or about 3-1/2 ounces per pound of dry fiber.
7. Heat the pot of water and add the washing soda before it boils. As the washing soda dissolves, add the soaked fiber and stir. Bring to a boil, and then turn down the heat and simmer.
8. Every half-hour while simmering, stir the fiber and test it for doneness. Take a piece of fiber, rinse it, and pull it in the direction of the plant's growth. If the fiber pulls apart easily, it's ready.

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9. Turn off the heat and remove the pot from the stove. Pour the cooked fiber through a strainer into a bucket (not down the drain yet), and rinse the fiber until the water runs clear. Make sure you remove all the washing soda at this point.
10. Dump the water outside, mix the plant juices with vinegar to neutralize the solution to avoid introducing toxin into the environment.
11. Beating the fiber. After the fiber is cooked, you'll have to beat it to further break down the material into the soft pulp that you'll use to make sheets of paper. Papermakers often hand-beat fibers, which generally results in the strongest paper. Others use large equipment specifically for papermaking. The easiest and quickest method at home is to use a nonfood blender. Add a handful of your cooked fiber to a blender (make sure it's fully hydrated; soak overnight if using stored fiber) and fill the blender about three-quarters full with water. Put the lid on, and then beat at a medium or high speed. If the blender sounds strained, check to make sure fiber isn't wrapped around the blades (don't forget to cut it into smaller pieces before cooking). The blender also might strain if one has added too much fiber. The length of blending time depends on the fiber. Try about 20 seconds at first and then increase in 20-second increments. One can detect pulp when the fibers don't clump and look very fine, almost cloud-like dispersed in the water. If strings of fibers are observed and looked clumpy, one need to blend it a bit longer.
12. : *Fill a vat with water and a few handfuls of pulp. With one hand, "hog" the vat—stir the pulp around and disperse it evenly in the water. Wet the mould and deckle. Vat is a tub larger than mould and deckle. Fill it with water and pulp and pull up sheets of paper from it. Use a large storage tub, dish tub, a freestanding plastic vat with a drain and plug on the bottom, or even an old secondhand sink.*
13. *13: Hold the mould and deckle together and insert it at a 45 degree angle into the water. Then, as the bottom of the vat is reached, turn it parallel to the water's surface and pull it up out of the water.*
14. *14: While pulling the mould and deckle, shake it gently back and forth to settle the fibers. Hold still and parallel for several seconds to let the water continue to drain out into the vat and let the fibers settle*
15. *Set the mould and deckle on the edge of the vat, lift the deckle, and sweep it swiftly over the paper and to the side (without dripping).*
16. *Hold the mould up over the vat to drain the rest of the water, holding until it only drips. The paper will stay secure on the mould.*
- 17: *Lay the mould, paper side down, onto a felt. Gently press all areas of the mould to help release the paper. Then, with pressure from one hand on one side of the mould, pull up by shifting pressure to the other side in a fluid motion. The paper will release onto the felt.*
- 18: *While the paper is still on the felt, sponge out excess water or place the paper in a press to squeeze out water (or wait until you've pulled multiple sheets to load them all into the press). Then, pick up the wet paper from the felt and place it on your drying surface, such as Plexiglas or corrugated cardboard. Roll smooth with a brayer or similar tool. The paper will be delicate but should hold together well at this point. Label it.* Observe plants transform from their original color to the hues and textures they'll take on after cooking, then into pulp, and finally to the look and feel of the final paper.
19. *When finished pulling paper, strain the water from the vat through a mesh-lined strainer into a bucket, and reserve the pulp for later use.*
20. *Construct a homemade press to squeeze out more water.*

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21. dry paper on Plexiglas or corrugated cardboard, which allows airflow.

#### **PHYSICAL ANALYSIS (JAFFUR AND JEETAH, 2019)**

##### 1. Grammage determination (Tappi T410 OM-08)

The grammage of a paper depends upon the size as well as the mass of the sheet. Different grammages of paper have different functions in the market such that paper having a small mass per unit area can be used as wrapping paper in the food industry or as toilet paper while those having a larger mass per unit area can be used as posters or cardboards [15]. Samples were cut in dimensions of 5 × 5 cm from different sheets of paper followed by weighing on an electronic weighing balance. The grammage was then determined as follows:

$$g = \frac{m}{A \times n}$$

where, g is the sheet grammage in g m<sup>-2</sup>, m is the mass of paper sample in g, A is the paper sample area in m<sup>2</sup>, and n is the number of samples.

##### 2. Thickness determination (Tappi T551 PM-12)

In order to determine the thickness of the paper specimen more accurately without compressing the paper surface, a tester from any authorized laboratory will be utilized. Specimen will be submitted to the said laboratory for thickness determination.

##### 3. TEXTURE EVALUATION

##### 4. GENERAL ACCEPTABILITY EVALUATION

##### 500) STATISTICAL ANALYSIS

Mean, percentage and analysis of variance/ chi-square could be possibly used for this study.

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**(14) Major Activities/Work plan** (Enumerate in chronological order the task to be undertaken. Use gantt chart.

| Research Objectives                             | Year 1 |     |     |     |      |     |     |     |     |     |     |     |  |
|-------------------------------------------------|--------|-----|-----|-----|------|-----|-----|-----|-----|-----|-----|-----|--|
|                                                 | May    | Jun | Jul | Aug | Sept | Oct | Nov | Dec | Jan | Feb | Mar | Apr |  |
| 1. Conceptualization/<br>Pre-trial/ observation |        |     |     |     |      |     |     |     |     |     |     |     |  |
| 2. Conduct of the<br>study                      |        |     |     |     |      |     |     |     |     |     |     |     |  |

| Research Objectives                                        | Year 2 |     |     |      |     |     |     |     |     |     |     |     |  |
|------------------------------------------------------------|--------|-----|-----|------|-----|-----|-----|-----|-----|-----|-----|-----|--|
|                                                            | Jun    | Jul | Aug | Sept | Oct | Nov | Dec | Jan | Feb | Mar | Apr | May |  |
| 3. Analysis of<br>data gathered<br>from<br>observation     |        |     |     |      |     |     |     |     |     |     |     |     |  |
| 3. Lay-outing<br>and Editing,<br>Presntation of<br>Results |        |     |     |      |     |     |     |     |     |     |     |     |  |

**(15) Expected Output** (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The study might produce paper out from recycled material, help reduce pollution and might help augment the income of small scale producers.

**(15) Technology Transfer Activity Design** (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

- processes and its product will be shared as part of extension activity

**(16) Targets Beneficiaries** (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

**(17) Personnel Requirement**

| Name                                                | Role<br>(Program,Project<br>, Study Leader) | Field of Specialization |
|-----------------------------------------------------|---------------------------------------------|-------------------------|
| Rhodora Ticar                                       | Study leader                                | General Science         |
| Percent time devoted to the Program/ Project/ Study |                                             |                         |
| Program/Project Leader (in hours):                  |                                             |                         |

|                                                                                   |                 |                                                |                |                          |
|-----------------------------------------------------------------------------------|-----------------|------------------------------------------------|----------------|--------------------------|
|  | Document Type:  | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Document Code  | <b>RES- RP-01</b>        |
|                                                                                   | Document Title: | <b>RESEARCH PROPOSAL</b>                       | Revision No.   | <b>00</b>                |
|                                                                                   |                 |                                                | Effective Date | <b>November 30, 2019</b> |
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**Program/Project Members (in hours):**

**(18) Budgetary Requirement**

| SOURCES OF EXPENSES                                                                                                                          | Quantity                               | Amount (PhP)     |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------|
| <b>A. Personnel Services</b> (For studies needed the Research Assistant, informant/s, guide)                                                 | <b>1 RESEARCH ASSISTANT (1 MONTHS)</b> | <b>13,000/MO</b> |
| <b>B. Maintenance and Other Operating Expenses (MOOE)</b> (modify and include only whichever is applicable)                                  |                                        |                  |
| -                                                                                                                                            |                                        |                  |
| -                                                                                                                                            |                                        |                  |
| - Professional fees (laboratory analysis and other related fees)                                                                             |                                        |                  |
| - Miscellaneous (for incidental expenses)                                                                                                    |                                        |                  |
| - Printing                                                                                                                                   |                                        |                  |
| <b>C. CAPITAL OUTLAY</b>                                                                                                                     |                                        |                  |
| <b>D. Purchase of Equipment /Materials</b>                                                                                                   |                                        |                  |
| 1. 1 pack rubber gloves                                                                                                                      | 1                                      | 100.00           |
| 2. Large Capacity Multifunctional Stainless Steel Cooking Pot                                                                                | 4 @ 2,500                              | 10,000.00        |
| 3. N95 face mask                                                                                                                             | 20 pcs                                 | 2,000.00         |
| 4. googles                                                                                                                                   | 2                                      | 700.00           |
| 5. nonfood blender                                                                                                                           | 4                                      | 8,000.00         |
| 6. big bucket or pail                                                                                                                        | 4                                      | 600.00           |
| 7. Large basin                                                                                                                               | 2                                      | 400              |
| 8. pot holders                                                                                                                               | 8                                      | 800.00           |
| 9. 4 pcs. Mesh strainer                                                                                                                      | 4                                      | 400.00           |
| 10. 4 nonfood, nonreactive stirring utensils                                                                                                 | 4                                      | 800.00           |
| 11. homemade press to squeeze out more water (2 pcs. 12inch x 12 inch x 2 inch plywood and 4 pcs 12inch x 2 inch x 3 inch lumber and screws) | 1                                      | 1,000.00         |
| 12. weighing scale                                                                                                                           | 1                                      | 2,000.00         |
| 13. Mould and deckle                                                                                                                         | 6                                      | 2,800.00         |
| 14. Hard felt cloth (1mm x 9 inches x 12 inches)                                                                                             | 12                                     | 240.00           |
| 15. Large rectangular plastic containier as Vat                                                                                              | 2                                      | 700.00           |
| 16. sponges                                                                                                                                  | 12                                     | 360.00           |
| 17. 4 inches rubber brayer roller                                                                                                            | 2                                      | 500.00           |
| 18. Caustic soda                                                                                                                             | 1kg                                    | 150.00           |
| <b>Total</b>                                                                                                                                 |                                        | <b>44,510.00</b> |

**(19) Literature Cited** Vinegar to neutralize the alkali solution

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|  | Document Type:  | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Document Code  | <b>RES- RP-01</b>           |
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Wolela, A.D. (2019). Extraction and Characterization of Natural Cellulose Fibers from Sanseveria Trifasciata Plant. *Advancements Bioequiv Availab*. 5(2). TTEFT.000609.2019. DOI: 10.31031/TTEFT.2019.05.000609.

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<https://www.intechopen.com/chapters/67083>

<https://www.britannica.com/science/cellulose>

## **(20) Capsule Curriculum Vitae (One-page brief C.V. only)**

**Name: RHODORA LACHICA TICAR**

**Position: Associate Professor V**

**Permanent Address: Paradise Street, Mambusao, Capiz**

**E-mail address: [rhoda.ticar@gmail.com](mailto:rhoda.ticar@gmail.com)**

### **Education:**

|                                                                                                                              |                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Doctor of Education, Major in Educational Management<br>Capiz State University, Mambusao, Capiz<br>October 2004 to July 2020 | 12 Units, MS Biology<br>University of the Philippines in the Visayas<br>June 1995 to March 1996               |
| 6 Units, Doctor of Science Education (Biological Science)<br>West Visayas State University<br>October 2006 to March 2007     | Bachelor of Secondary Education, Major in Biology<br>West Visayas State University, Iloilo City<br>March 1995 |

|                                                                                   |                 |                                                |                |                             |
|-----------------------------------------------------------------------------------|-----------------|------------------------------------------------|----------------|-----------------------------|
|  | Document Type:  | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Document Code  | <b>RES- RP-01</b>           |
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Master of Arts in Education, Major in General Science  
West Visayas State University, Iloilo City  
October 1996 to October 2004

**Eligibility: Professional Board Examination (79.91%)**

**Work Experience:**

**Faculty**

Capiz State University (formerly Panay State Polytechnic College)  
Mambusao, Capiz  
September 1996 to present

**Research Chairman**

Capiz State University- Mambusao Campus  
2011 to June 2019

**Intellectual Property Rights Chairman**

Capiz State University- Mambusao Campus  
2015 to July 2022

**Extension Chairman**

Capiz State University- Mambusao Campus  
July 2022 to date